Synopsis

Clinical Risk Assessment for Type-2 Diabetes

Problem Id:07

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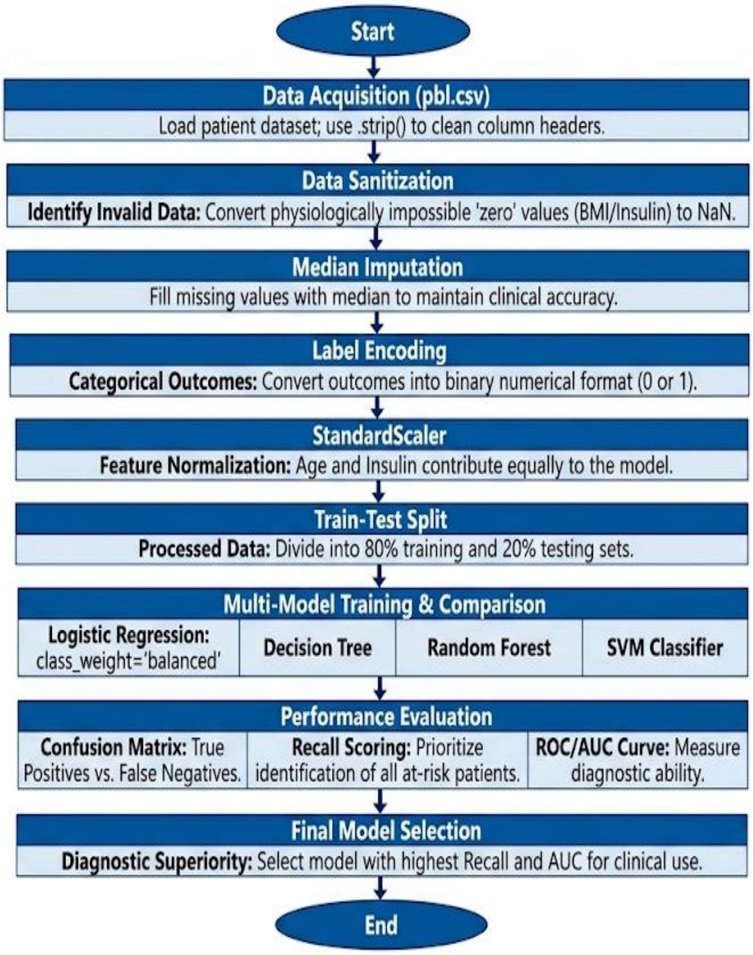
- **Problem Description:**

  Type-2 Diabetes is one of the most common chronic diseases worldwide, and early prediction is critical in preventing severe complications such as heart disease, kidney failure, and nerve damage. The primary objective of this project is to develop a machine learning model that predicts the clinical risk of Type-2 Diabetes based on key health parameters: Age, Body Mass Index (BMI), and Insulin Levels.
- Data Acquisition & Cleaning: The project utilizes a clinical dataset (pbl.csv) to simulate real patient scenarios. To ensure data integrity, the system cleans column headers and identifies physiologically impossible "zero" values in BMI and Insulin. These are handled via Median Imputation, ensuring the model is not skewed by missing data.
- Preprocessing & Normalization: Categorical risk outcomes are converted into numerical form using Label Encoding. Furthermore, StandardScaler is applied to normalize the data. This is essential because parameters like Insulin often have much higher numerical ranges than Age; scaling ensures all features contribute equally to the final prediction.
- Comparative Modeling: While the core logic utilizes a Logistic Regression classifier with "balanced" class weights, the project further extends to a comparative analysis of Decision Trees, Random Forests, and SVM. This allows for the selection of the most robust model for health diagnostics.
- Evaluation Strategy: Performance is measured using a Confusion Matrix and ROC Curves. We place a strategic focus on the Recall Score, as correctly identifying at-risk patients (avoiding False Negatives) is more vital in medical applications than minimizing false alarms . Ultimately, this project demonstrates how machine learning serves as a reliable decision support system for healthcare professionals.

- **Steps of Implementation:**



This flowchart outlines the steps involved in the process, from data collection to model evaluation.

- **Expected Modules/ Libraries to be used :**

- **NumPy**
  Used for numerical operations and generating synthetic numerical health data.
- **Pandas**
  Used for dataset creation, data manipulation, and structured data handling.
- **Scikit-learn(sklearn)**
  Used for preprocessing (LabelEncoder, StandardScaler), Logistic Regression model, and evaluation metrics.
- **MatplotlibSeaborn**
  Used for plotting graphs and Confusion Matrix visualization.
- **Random**
  Used for generating synthetic data within medical ranges.

- **Expected Evaluation metrics and Plots to be used:**

- **ConfusionMatrix**
  The Confusion Matrix will be used to evaluate the classification performance by showing

True Positive, True Negative, False Positive, and False Negative predictions. It helps in understanding how well the model distinguishes between risk and non-risk patients.

- **RecallScore**
  Recall is important in this project because it measures how many actual diabetic risk patients are correctly identified. In medical applications, missing a positive case can be dangerous, so Recall is an important metric.
- **AccuracyScore**
  Accuracy will be used to measure the overall correctness of the model by calculating the ratio of correct predictions to total predictions.
- **HeatmapPlot**
  A heatmap visualization of the Confusion Matrix will be used to make the evaluation more interpretable and visually clear.
- **ROC Curve** A plot used to measure the area under the curve for model reliability.
- **FeatureDistributionPlot**
  Basic plots such as histograms may be used to visualize the distribution of Age, BMI, and Insulin levels in the synthetic dataset.

- **Required topics from the subject:**

- **Arrays**
  Arrays are used to store multiple values of health parameters such as Age, BMI, and Insulin levels. In machine learning, datasets are internally handled as arrays or matrices.
- **Strings**
  Strings are used to handle categorical data such as risk labels (High Risk / Low Risk). These values are later converted into numerical format using encoding techniques.
- **FileHandling**
  File handling concepts are useful for reading and storing datasets in CSV format. It allows saving the generated synthetic data and reusing it for model training and testing.
- **Functions**
  Functions help in modular programming by separating different stages such as data preprocessing, model training, and evaluation. This improves code readability and reusability.
- **Libraries/Modules**
  Understanding libraries is important because machine learning depends on external modules like NumPy, Pandas, and sklearn which provide predefined functions for efficient implementation.

- **Platform Required:**

Visual studio

- **Books and Link Sources:**

Online Link Sources:

1. Scikit-learnDocumentation
   https://scikit-learn.org

2. PandasDocumentation
   https://pandas.pydata.org

3. NumPyDocumentation
https://numpy.org
4. MatplotlibDocumentation
https://matplotlib.org